

AMAL JYOTHI COLLEGE OF ENGINEERING (AUTONOMOUS)

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Research Project / Software Project Part 1 – 24MCAR295 | MCA 2025-27-S3

Project Synopsis / Proposal

MediSense: An AI- Driven Patient Medication Non-Adherence and Chronic Health Risk Mitigation Portal

1. Introduction

In modern healthcare management, patient safety and precise medication adherence remain critical challenges. Healthcare providers often face difficulties in tracking complex patient prescription regimens, monitoring dosage compliance, and identifying adverse drug interactions in real time. MediSense is an advanced, AI-driven Clinical Decision Support System (CDSS) and Patient Intelligence Platform designed to streamline medical record management, automate risk score calculations, and provide real-time alerts for healthcare professionals and patients.

2. Problem Definition

Existing hospital management and patient logging workflows rely heavily on manual data entry or fragmented systems. This introduces significant operational bottlenecks, delays emergency risk identification, and increases the likelihood of medication non-adherence and adverse drug events. Furthermore, most clinical systems lack interactive, real-time feedback loops and intelligent risk scoring engines that can analyze historical intake patterns alongside clinical metrics to assist doctors during diagnosis.

3. Objectives

- To design and develop a responsive, high-performance web dashboard with real-time UI updates for clinical and patient management.
- To implement a robust Role-Based Access Control (RBAC) system for Patients, Doctors, and Administrators.
- To engineer a Patient Intake Logging & Adherence Tracker with dynamic notification mechanisms.
- To develop an AI-assisted Risk Calculation Engine that computes real-time patient risk scores based on adherence logs and health metrics.
- To validate the system's accuracy and reliability through multi-user clinical simulation and live environment hosting.

4. Literature Survey / Existing Systems

A survey of existing systems in the healthcare management and EMR space reveals several functional gaps:

- Traditional EHR/EMR Systems (Epic, Cerner): Focus heavily on clinical data storage and administrative billing, but lack lightweight, patient-facing adherence automation and real-time interactive risk scoring.
- Medication Tracker Apps (MyTherapy, Medisafe): Standalone mobile applications offering simple reminders, but lack integration with clinical doctor dashboards and automated risk decision support systems.
- Standalone Clinical Decision Support Tools: Complex, isolated algorithmic calculators that operate independently of actual patient daily intake logs and prescription history.

MediSense bridges these gaps by combining patient-side intake tracking with doctor-side AI decision support in a single, integrated full-stack solution.

5. Proposed System

MediSense utilizes Django for robust backend ORM management and HTMX with Tailwind CSS for dynamic, SPA-like client interactivity without the overhead of heavy frontend frameworks. The platform integrates a custom Python Risk Analytics Engine and a PostgreSQL persistence store to process intake logs, generate automated risk alerts, and provide actionable clinical insights to doctors via interactive dashboards.

6.Scope of the Project

The project is structured across two distinct academic phases:

- **[MINI PROJECT - Phase 1]: Core Foundation & Intake Management** Focuses on architectural setup, secure multi-role authentication (RBAC), database design, patient prescription logging, basic adherence tracking, and core doctor/patient dashboard interfaces.
- **[MAIN PROJECT - Phase 2]: AI Analytics, Intelligence & Live Deployment** Extends Phase 1 with the AI Risk Engine for real-time risk scoring, automated clinical alert generation, advanced analytics visualization, multi-tenant patient onboarding, and cloud hosting validation.

7. System Requirements

Category	Requirement
Backend	Django Framework, FAST API
Frontend	HTMX, Tailwind CSS, Alpine.js, Chart.js
Database and Caching	PostgreSQL
AI / Analytics Engine	Python (NumPy, Pandas, Scikit-learn / Rule-based Risk Engine)
Containerization & Deployment	Docker, Nginx, Gunicorn
Development Tools	Git, GitHub, VS Code, Postman, PgAdmin

8. Modules

- Authentication & Role-Based Access Control (RBAC)
- Patient Management & Profile Center
- Prescription & Dosage Scheduler
- Intake Logging & Adherence Tracker (*Mini Project*)
- AI Risk Calculation & Analytics Engine (*Main Project*)
- Doctor Clinical Decision Dashboard
- Real-time Notification & Alert Center
- Audit Trail & System Log Management

9. Methodology / Work Plan

Phase	Duration	Deliverable
Requirement Gathering & Design	Weeks 1–4	Abstract, Synopsis, UML diagrams, DB design, UI design
Phase 1 Development	Weeks 5–10	Core Platform: Auth, RBAC, Patient Intake Logging, Basic Dashboards
Interim Evaluation	Weeks 9–10	60% completion, literature review, tool evaluation
Phase 1 Completion	Weeks 11–12	Full Mini Project Deliverable: Complete CRUD & Adherence Tracking
Phase 2 Development	Weeks 13–15	AI Risk Engine integration, Real-time Alerts, Doctor CDSS Dashboard, Cloud Hosting
Final Evaluation	Weeks 16–17	Testing, report, presentation, final internal evaluation

10. Expected Outcome

MediSense will deliver a unified, high-reliability clinical decision platform that drastically reduces medication tracking errors, automates risk prioritization for doctors, and improves patient outcome tracking through intelligent analytics.

11. References

- Django Software Foundation, "Django Documentation and Guidelines," djangoproject.com.
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- World Health Organization (WHO), "Adherence to Long-Term Therapies: Evidence for Action," WHO Guidelines.
- Scikit-learn Developers, "Machine Learning in Python - Scikit-learn Documentation," scikit-learn.org.

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